

WestQuant Hamiltonian Representation Explorer (Project B)

Slide 1 — Project Overview

Problem A mathematical optimization problem does not determine a unique useful Hamiltonian representation. The penalty strength, varia

Approach Generate multiple mathematically valid Hamiltonian representations (scaling, permutation, bit-complement, MWIS penalty family

Results For MWIS, penalty strengths U in $[5.5, 22.2]$ all yield GROUND_STATE_EQUIVALENT Hamiltonians. The Hamiltonian main effect is n

Importance The Hamiltonian representation choice matters less than its interaction with the embedding. Representation itself is an optimiz

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Slide 2 — QoolQit Experience

Positive aspects: QoolQit's `DataGraph.from_matrix` and `interaction_matrix()` made it easy to connect logical Hamiltonians to physical interactions.

Challenges: Native Rydberg interactions are repulsive ($J > 0$), so Hamiltonians with negative pair interactions (e.g., after bit-complement) are not directly representable.

Workflow: The five-class equivalence taxonomy forced mathematical rigor. Exhaustive verification for $n \leq 16$ ensured no false equivalence.